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How would you go about calculating a person's carbon footprint impact from using generative AI?

Posted by u/Sharp-Appointment795 • • 0 points • 0 comments

Here's my wild guess with very little in terms of references:

Average daily use 30,000 tokens

Throughput of a single H100 card (the standard) is let's say 150 tokens per second (had trouble finding references)

Power consumption of H100: 500w

Power consumption of datacenter equipment to support H100 like cooling, networking and backups +20%

Average carbon footprint of a US datacenter: 0.1 tons of CO2 per kilowatt hour (kWh)

$30,000 \text{ tokens} / 150 \text{ tps} = 67 \text{ seconds of generation}$

Total datacenter power consumption of H100: $500 \text{w} \times 120\% = 600 \text{w}$

$67 \text{ seconds of H100 power use } 600 \text{w} / 1200 \text{ seconds in an hour} \times 60 = 33.5 \text{ Watts}$

$33.5 \text{ Watts} \times 30 \text{ days} = 1,005 \text{ Watts}$

If I round to 1,000 Watts that's 1kWh or 0.1 tons of CO2

How would you calculate it? Got reliable data sources with different numbers?

Thanks!

Comments